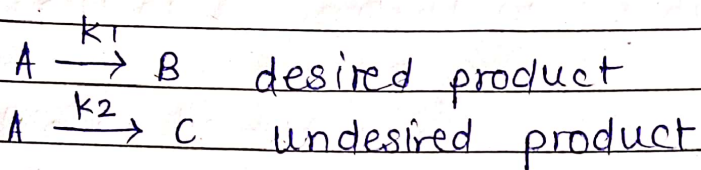


Q. Explain series, parallel and independent reactions with suitable examples.

S. Derive relationship for C_F in terms of ψ for plug flow reactor in case of parallel reaction.

Parallel Reaction Setup:

Consider set of parallel reactions of form:



Let

x = conversion of A

- $r_1 = k_1 C_A$ rate of formation of B
- $r_2 = k_2 C_A$ rate of formation of C
- $\psi = \frac{k_2}{k_1}$ the selectivity ratio

define

$$C_{RF} = \frac{r_1}{r_1 + r_2} = \frac{k_1 C_A}{k_1 C_A + k_2 C_A} = \frac{1}{1 + \psi}$$

- molar balances in PFR

PFR eqⁿ

$$\frac{dF_A}{dV} = -r_A = -(r_1 + r_2) = -(k_1 + k_2) C_A = -k_1 (1 + \psi) C_A$$

Using conversion x .

$$C_A = C_{A0} (1 - x) \quad \text{and}$$

$$F_A = F_{A0} (1 - x)$$

$$\frac{dx}{dV} = \frac{-1}{F_{A0}} \cdot \frac{dF_A}{dV} = \frac{k_1 (1 + \psi) C_A}{F_{A0}} = \frac{k_1 (1 + \psi) C_{A0} (1 - x)}{F_{A0}}$$

Now integrate from $X=0$ to X and $V=0$ to V

$$\int_0^X \frac{dx}{1-X} = \frac{k_1(1+\psi)C_{A0}}{F_{A0}} \int_0^V dv = -\ln(1-X) = \frac{k_1(1+\psi)C_{A0}}{F_{A0}} V$$

• Define CRF (Yield fraction of B)

let

$$- Y_B = \frac{\text{moles of B formed}}{\text{mole of A fed}}$$

$$- Y_C = \frac{\text{moles of C formed}}{\text{moles of A fed}}$$

$$- X = Y_B + Y_C$$

Then

$$CRF = \frac{Y_B}{X}$$

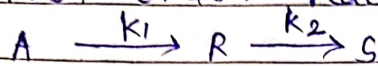
using rate expression

$$\frac{dY_B}{dx} = \frac{r_1}{r_1+r_2} = \frac{1}{1+\psi} \Rightarrow Y_B = \int_0^X \frac{1}{1+\psi} dx = \frac{X}{1+\psi}$$

$$\therefore CRF = \frac{Y_B}{X} = \frac{X}{(1+\psi)X} = \frac{1}{1+\psi}$$

$$CRF = \frac{1}{1+\psi}$$

Q. Explain quantitative treatment of mixed flow reactor for first order reaction in series.
 → Consider irreversible unimolecular type first order reaction in series.



Occurring in mixed flow reactor

The steady-state material balance Component A.

Input of A = output of A + disappear A by reaction.

$$F_{A0} = F_A + (-r_A) V$$

$$V_0 C_{A0} = V C_A + (-r_A) V$$

$$V C_{A0} = V C_A + (-r_A) V \quad \text{for constant } V_0 = V$$

$$-r_A = k_1 C_A$$

$$V C_{A0} = V C_A + k_1 C_A V \quad \text{--- eq}^n 1)$$

$$\text{we have } \tau_m = \bar{t} = V \quad \text{--- 2)}$$

divide eqⁿ 1) and combine with 2)

$$C_{A0} = C_A + k_1 C_A (\tau_m)$$

$$C_{A0} = C_A (1 + k_1 \tau_m)$$

$$\frac{C_A}{C_{A0}} = \frac{1}{1 + k_1 \tau_m} \quad \text{eq}^n 3)$$

for component R:

Input of R + Generation of R = output of R

$$V C_{R0} + (r_R) V = V C_R \quad \text{--- eq}^n 3)$$

$$V C_{R0} + (k_1 C_A - k_2 C_R) V = V C_R$$

$$C_{R0} = 0$$

$$\therefore (k_1 C_A - k_2 C_R) V = V C_R$$

$$k_1 C_A V = k_2 C_R V = V C_R$$

$$C_R = k_1 C_A \frac{V}{V} - k_2 C_R \frac{V}{V}$$

$$\text{put } \tau = \frac{V}{V}$$

$$C_R = k_1 C_A \tau_m - k_2 C_R \tau_m$$

$$C_R (1 + k_2 \tau_m) = k_1 C_A \tau_m \quad \text{--- eq}^n 4)$$

$$\text{from eq}^n 3) \quad C_A = \frac{C_{A0}}{(1 + k_1 \tau_m)}$$

$$C_R (1 + k_2 \tau_m) = \frac{k_1 C_{A0} \tau_m}{(1 + k_1 \tau_m)}$$

$$C_R = \frac{k_1 C_{A0} \tau_m}{(1 + k_1 \tau_m)(1 + k_2 \tau_m)} \quad \text{eq}^n 5)$$

$$\frac{C_R}{C_{A0}} = \frac{k_1 \tau_m}{(1 + k_1 \tau_m)(1 + k_2 \tau_m)} \quad \text{eq}^n 6)$$

$$C_A + C_R + C_S = C_{A0}$$

$$C_S = C_{A0} - C_A - C_R$$

$$C_S = C_{A0} - \frac{C_{A0}}{(1 + k_1 \tau_m)} - \frac{k_1 C_{A0} \tau_m}{(1 + k_1 \tau_m)(1 + k_2 \tau_m)}$$

$$\frac{C_S}{C_{A0}} = \frac{k_1 k_2 \tau_m^2}{(1 + k_1 \tau_m)(1 + k_2 \tau_m)} \quad \text{eq}^n 7)$$

$$C_R = \frac{C_{A0} k_1 \tau_m}{(1 + k_1 \tau_m)(1 + k_2 \tau_m)}$$

$$\frac{dC_R}{d\tau_m} = 0 = \frac{C_{A0} k_1 (1 + k_1 \tau_m)(1 + k_2 \tau_m) - (C_{A0} k_1 \tau_m [k_1 (1 + k_2 \tau_m) + k_2 (1 + k_1 \tau_m)])}{(1 + k_2 \tau_m)^2 + (1 + k_2 \tau_m)^2}$$

$$C_{A0} k_1 (1 + k_1 \tau_m)(1 + k_2 \tau_m) - C_{A0} k_1 \tau_m [k_1 (1 + k_2 \tau_m) + k_2 (1 + k_1 \tau_m)] = 0$$

$$C_{A0} k_1 (1 + k_1 \tau_m)(1 + k_2 \tau_m) = C_{A0} k_1 \tau_m [k_1 (1 + k_2 \tau_m) + k_2 (1 + k_1 \tau_m)] = 1$$

$$1 + k_2 \tau_m + k_1 \tau_m + k_1 k_2 \tau_m^2 = k_1 \tau_m + k_1 k_2 \tau_m^2 + k_2 \tau_m + k_1 k_2 \tau_m^2$$

$$1 = k_1 k_2 \tau_m^2$$

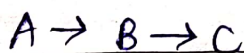
$$\tau_m^2 = \frac{1}{k_1 k_2}$$

$$\tau_{m \text{ opt}} = \frac{1}{\sqrt{k_1 k_2}}$$

3. Explain various type of multiple reactions with suitable examples.

→ 1. Series Reactions (consecutive reactions)
One product of reaction becomes a reactant in next

General form



Q. Explain terms used to characterize parallel reactions

→ Rate of Reaction -

The speed at which each reaction occurs.

- $A \rightarrow B$ (desired) at rate $r_1 = k_1 [A]^n$
- $A \rightarrow C$ (undesired) at rate $r_2 = k_2 [A]^m$

where :

k_1, k_2 = rate constant

n, m = reaction order

$[A]$ = Concentratⁿ of reactant A.

2. Selectivity (S)

- Describes how selectively desired product is formed over undesired product

$$S = \frac{\text{moles of desired product formed}}{\text{moles of undesired product formed}}$$

$$S = r_1 / r_2$$

3. Yield (Y)

- fractⁿ of reactant converted into desired product

$$Y = \frac{\text{moles of desired product formed}}{\text{moles of reactant consumed}}$$

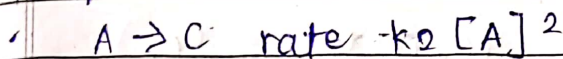
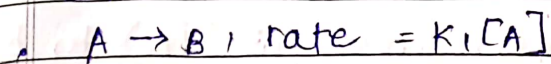
4. Conversion (X)

- Fractⁿ or percentage of reactant that is reacted.

$$X = \frac{\text{moles of A reacted}}{\text{initial moles of A}}$$

6. Reactⁿ order - (n, m)

- How rate of each parallel reactⁿ depend on conc. of reactant



6. Rate constant (k)

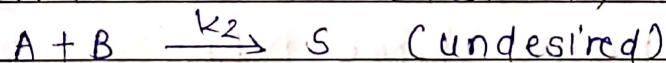
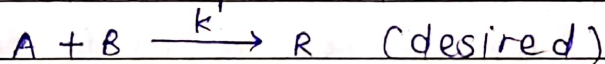
- temp- dependent coefficient indicating intrinsic speed of reactⁿ. Diff. reactions in parallel have different rate constant.

Arrhenius eqⁿ

$$k = A \cdot e^{(-E_a/RT)}$$

Q. Give quantitative treatment of product distribution and reactor size for parallel reaction.

→ consider parallel reactⁿ



rate of formation of R is

$$r_R = k_1 C_A^\alpha \text{ — eqⁿ 1)}$$

rate of formatⁿ of S is

$$r_S = k_2 C_A^\beta \text{ — 2)}$$

divide eqⁿ 1) and 2)

$$\frac{r_R}{r_S} = \frac{k_1}{k_2} C_A^{\alpha-\beta} \text{ — 3)}$$

eqⁿ 3) gives molecular relative rate of formation R and S. The ratio $\frac{r_R}{r_S}$ called

rate selectivity s .

$$S = \frac{r_R}{r_S} = \frac{k_1}{k_2} C_A^{\alpha-\beta}$$

We want rate of formation $R(r_R)$ to be high w.r.t rate of formatⁿ $S(r_S)$

ratio $\frac{r_R}{r_S}$ to be as high as possible.

Case 1) $\alpha > \beta$ $n_{\text{desire}} > n_{\text{undesire}}$

High reactant conc. is desirable as increase with R ratio or maximize selectivity. A batch PFR fav. to R which require minimum reactor size

2) $\alpha < \beta$ $n_{\text{desire}} < n_{\text{undesire}}$

Low reactant conc is desire as it maxi selectivity require large MFR.

3) $\alpha = \beta$ $n_{\text{desire}} = n_{\text{undesire}}$

product distribution fixed by k_1/k_2 alone. Conc. level does not affected by selectivity is unaffected by type reactor.

- Quantitative treatment for product distribution & reactor size.

- IF kinetics of individual reactⁿ are known then possible determine product distribution and reactor size needed.